

7 CONCLUSION

As LLMs evolve from text completers to autonomous agents, the standard chat interface becomes inadequate for tasks requiring persistent hidden state. We formalize this setting as Private State Interactive Tasks (PSITs) and show theoretically that agents conditioned only on public history cannot simultaneously keep the secret unresolved in the public transcript and remain consistent with a fixed hidden state. Empirically, we find that standard LLMs and retrieval-based baselines fail these tasks, exhibiting either state loss or information leakage. Our results highlight Private Working Memory as a necessary architectural component for PSITs. By explicitly maintaining private state, agents can separate internal reasoning from public actions and recover consistency with a fixed hidden state. Future research should explore additional interactive tasks where private working memory can improve an agent’s ability to manage complex internal states.

8 ACKNOWLEDGEMENTS

Sarath Chandar is supported by the Canada CIFAR AI Chairs program, the Canada Research Chair in Lifelong Machine Learning, and the NSERC Discovery Grant. Experiments were conducted using computational resources provided by Mila Quebec AI Institute.

REFERENCES

- Alaa Abd-Alrazaq, Rawan AlSaad, Dari Alhuwail, Arfan Ahmed, Padraig Mark Healy, Syed Latifi, Sarah Aziz, Rafat Damseh, Sadam Alabed Alrazak, and Javaid Sheikh. Large language models in medical education: opportunities, challenges, and future directions. *JMIR medical education*, 9(1):e48291, 2023.
- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Sandhini Agarwal, Lama Ahmad, Jason Ai, Sam Altman, Andy Applebaum, Edwin Arbus, Rahul K Arora, Yu Bai, Bowen Baker, Haiming Bao, et al. gpt-oss-120b & gpt-oss-20b model card. *arXiv preprint arXiv:2508.10925*, 2025.
- Anthropic. Building effective agents. 2024. URL <https://www.anthropic.com/engineering/building-effective-agents>. Engineering blog post on Anthropic’s website.
- Richard C Atkinson and Richard M Shiffrin. Human memory: A proposed system and its control processes. In *Psychology of learning and motivation*, volume 2, pp. 89–195. Elsevier, 1968.
- Alan David Baddeley. Working memory. *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, 302(1110):311–324, 1983.
- Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George Bm Van Den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. Improving language models by retrieving from trillions of tokens. In *International conference on machine learning*, pp. 2206–2240. PMLR, 2022.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- Harrison Chase et al. Langchain: A framework for developing applications with large language models. <https://github.com/langchain-ai/langchain>, 2022. Software.
- Aaron Chatterji, Thomas Cunningham, David J Deming, Zoe Hitzig, Christopher Ong, Carl Yan Shan, and Kevin Wadman. How people use chatgpt. Technical report, National Bureau of Economic Research, 2025.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.
- Brown Ebouky, Andrea Bartezzaghi, and Mattia Rigotti. Eliciting reasoning in language models with cognitive tools. *arXiv preprint arXiv:2506.12115*, 2025.

- Jizhan Fang, Xinle Deng, Haoming Xu, Ziyang Jiang, Yuqi Tang, Ziwen Xu, Shumin Deng, Yunzhi Yao, Mengru Wang, Shuofei Qiao, et al. Lightmem: Lightweight and efficient memory-augmented generation. *arXiv preprint arXiv:2510.18866*, 2025.
- Arsene Fansi Tchango, Rishab Goel, Zhi Wen, Julien Martel, and Joumana Ghosn. Ddxplus: A new dataset for automatic medical diagnosis. *Advances in neural information processing systems*, 35:31306–31318, 2022.
- GitHub. vscode-copilot-chat. <https://github.com/microsoft/vscode-copilot-chat>, 2025. Accessed: 2025-08-13.
- Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. Hipporag: Neurobiologically inspired long-term memory for large language models. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- Aspen K Hopkins, Alex Renda, and Michael Carbin. Can llms generate random numbers? evaluating llm sampling in controlled domains. In *ICML 2023 workshop: sampling and optimization in discrete space*, 2023.
- Mengkang Hu, Tianxing Chen, Qiguang Chen, Yao Mu, Wenqi Shao, and Ping Luo. Hiagent: Hierarchical working memory management for solving long-horizon agent tasks with large language model. *arXiv preprint arXiv:2408.09559*, 2024.
- Pengbo Hu and Xiang Ying. Unified mind model: Reimagining autonomous agents in the llm era. *arXiv preprint arXiv:2503.03459*, 2025.
- Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating memory in llm agents via incremental multi-turn interactions, 2025. URL <https://arxiv.org/abs/2507.05257>.
- Shenyang Huang, Ali Parviz, Emma Kondrup, Zachary Yang, Zifeng Ding, Michael Bronstein, Reihaneh Rabbany, and Guillaume Rabusseau. Are large language models good temporal graph learners? *arXiv preprint arXiv:2506.05393*, 2025a.
- Yuxuan Huang, Yihang Chen, Haozheng Zhang, Kang Li, Huichi Zhou, Meng Fang, Linyi Yang, Xiaoguang Li, Lifeng Shang, Songcen Xu, Jianye Hao, Kun Shao, and Jun Wang. Deep research agents: A systematic examination and roadmap. *arXiv preprint*, 2025b. URL <https://arxiv.org/abs/2506.18096>.
- Hugging Face. Hugging face transformers: Chat templating. https://huggingface.co/docs/transformers/chat_templating, 2025. Accessed: 2025-09-13.
- Saurav Jha, Maryam Hashemzadeh, Ali Saheb Pasand, Ali Parviz, Min-Joong Lee, and Boris Knyazev. Ream: Merging improves pruning of experts in llms. *arXiv preprint arXiv:2604.04356*, 2026.
- David Kadavy. *Digital Zettelkasten: Principles, Methods, & Examples*. Kadavy, Inc., 2021.
- Adam Tauman Kalai, Ofir Nachum, Santosh S Vempala, and Edwin Zhang. Why language models hallucinate. *arXiv preprint arXiv:2509.04664*, 2025.
- Jiazheng Kang, Mingming Ji, Zhe Zhao, and Ting Bai. Memory os of ai agent. *arXiv preprint arXiv:2506.06326*, 2025.
- Mustafa O. Karabag, Jan Sobotka, and Ufuk Topcu. Do llms strategically reveal, conceal, and infer information? a theoretical and empirical analysis in the chameleon game, 2025. URL <https://arxiv.org/abs/2501.19398>.
- Oliver Kramer and Jill Baumann. Unlocking structured thinking in language models with cognitive prompting. *arXiv preprint arXiv:2410.02953*, 2024.
- LangChain. LangGraph: Build stateful, multi-actor applications with LLMs. <https://www.langchain.com/langgraph>, 2025. Accessed on November 25, 2025.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474, 2020.
- Zhiyu Li, Shichao Song, Hanyu Wang, Simin Niu, Ding Chen, Jiawei Yang, Chenyang Xi, Huayi Lai, Jihao Zhao, Yezhaohui Wang, et al. Memos: An operating system for memory-augmented generation (mag) in large language models. *arXiv preprint arXiv:2505.22101*, 2025.

- Jerry Liu et al. Llamaindex (formerly gpt index): A data framework for building llm applications. https://github.com/run-llama/llama_index, 2023. Software.
- Andreas Madsen, Sarath Chandar, and Siva Reddy. Are self-explanations from large language models faithful? In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 295–337, 2024.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of llm agents. *arXiv preprint arXiv:2402.17753*, 2024.
- Harry Mayne, Ryan Othniel Kearns, Yushi Yang, Andrew M Bean, Eoin D Delaney, Chris Russell, and Adam Mahdi. Llms don’t know their own decision boundaries: The unreliability of self-generated counterfactual explanations. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 24172–24197, 2025.
- João Moura et al. Crewai: A framework for orchestrating role-playing autonomous ai agents. <https://github.com/crewAIInc/crewAI>, 2024. Software.
- Boye Niu, Yiliao Song, Kai Lian, Yifan Shen, Yu Yao, Kun Zhang, and Tongliang Liu. Flow: A modular approach to automated agentic workflow generation. *arXiv e-prints*, pp. arXiv–2501, 2025.
- OpenAI. Handling raw chain-of-thought for GPT-OSS models. <https://cookbook.openai.com/articles/gpt-oss/handle-raw-cot>, 2025a. Accessed: 2025-11-24.
- OpenAI. Reasoning. <https://platform.openai.com/docs/guides/reasoning>, 2025b. Accessed: 2025-11-24.
- Charles Packer, Vivian Fang, Shishir G Patil, Kevin Lin, Sarah Wooders, and Joseph E Gonzalez. Memgpt: Towards llms as operating systems. *CoRR*, 2023.
- Zhuoshi Pan, Qianhui Wu, Huiqiang Jiang, Menglin Xia, Xufang Luo, Jue Zhang, Qingwei Lin, Victor Rühle, Yuqing Yang, Chin-Yew Lin, et al. LlmLingua-2: Data distillation for efficient and faithful task-agnostic prompt compression. *arXiv preprint arXiv:2403.12968*, 2024.
- Letitia Parcalabescu and Anette Frank. On measuring faithfulness or self-consistency of natural language explanations. In *ACL (1)*, 2024.
- Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*, pp. 1–22, 2023.
- Ali Parviz and Yuichi Yoshida. Nonlinear laplacians improve signed-directed graph learning. In *New Perspectives in Graph Machine Learning*, 2025a. URL <https://openreview.net/forum?id=3CcP3VI6LW>.
- Ali Parviz and Yuichi Yoshida. Nonlinear laplacians improve signed-directed graph learning. In *New Perspectives in Graph Machine Learning*, 2025b. URL <https://openreview.net/forum?id=3CcP3VI6LW>.
- Letian Peng and Jingbo Shang. Codifying character logic in role-playing, 2025. URL <https://arxiv.org/abs/2505.07705>.
- Nikolai Rozanov and Marek Rei. Stateact: Enhancing llm base agents via self-prompting and state-tracking. *arXiv preprint arXiv:2410.02810*, 2024.
- Ranjan Sapkota, Konstantinos I Roumeliotis, and Manoj Karkee. Ai agents vs. agentic ai: A conceptual taxonomy, applications and challenges. *arXiv preprint arXiv:2505.10468*, 2025.
- Amin Sghaier, Ali Parviz, and Alexia Jolicoeur-Martineau. Probabilistic tiny recursive model. *arXiv preprint arXiv:2605.19943*, 2026.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: language agents with verbal reinforcement learning. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 8634–8652. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/1b44b878bb782e6954cd888628510e90-Paper-Conference.pdf.
- Robyn Speer. rspeer/wordfreq: v3.0. September 2022. doi: 10.5281/zenodo.7199437. URL <https://doi.org/10.5281/zenodo.7199437>.